



DENTAL COMPOSITE RESTORATIONS

LEARNING OUTCOMES

- State and explain the basic composition of composites.
- Outline the Classification of composites.
- Describe the Polymerization reaction of composite.



INTRODUCTION



- The search for an ideal esthetic material for restoring teeth has resulted in significant improvements in esthetic restorative materials
- Composites are presently the most popular tooth-colored materials.
- Composite and Acid etch technique represent two major advances in restorative dentistry.



- **M.G.Bunocore** - 1955 – Introduced acid-etching technique
- In 1962, **Dr. R.L. Bowen** developed a polymer based on methacrylate chemistry.
- This polymer was generally known as **Bis-GMA** or Bowen resin, was made up from the combination of bisphenol – A and glycidyl methacrylate.

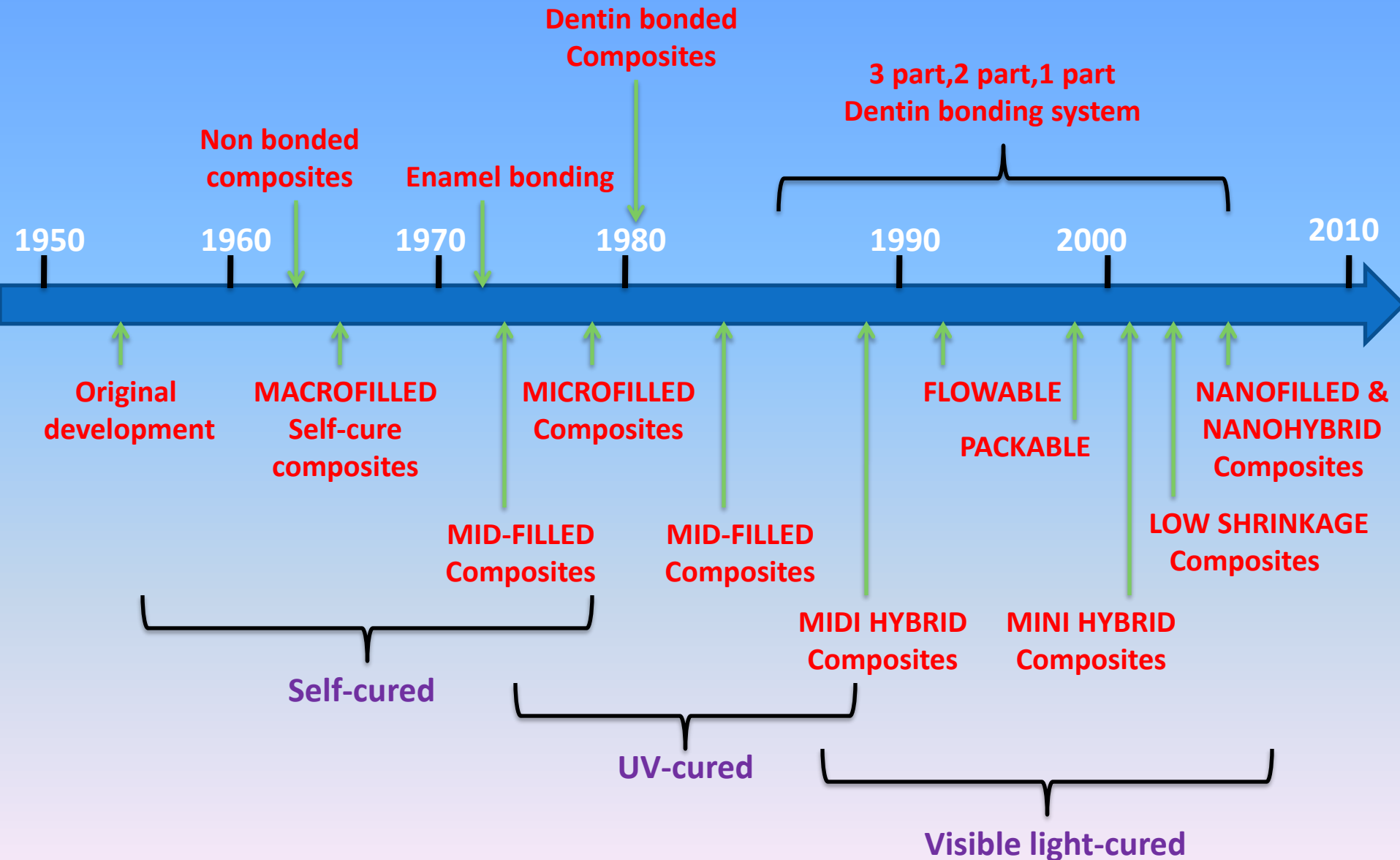


Definition of Composite resin –

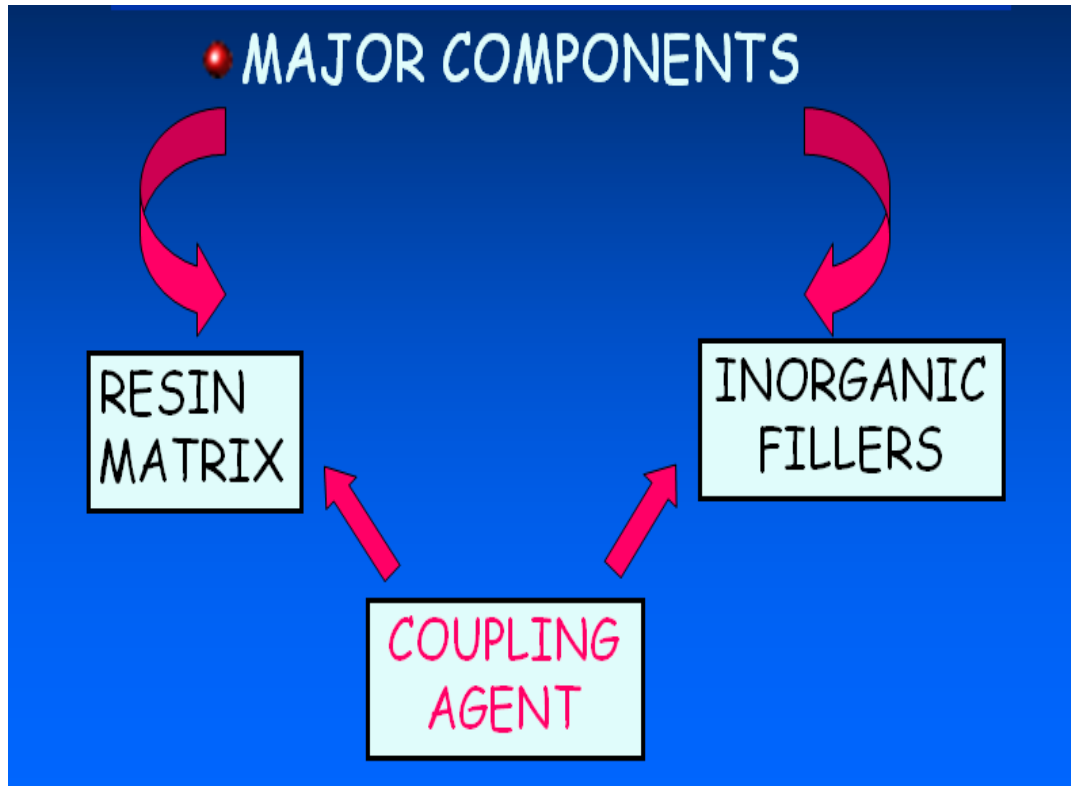
It is a 3 dimensional combination of two or more chemically different materials with a distinct interface between them, in combination the properties are superior to those of individual components.

Composite refinements

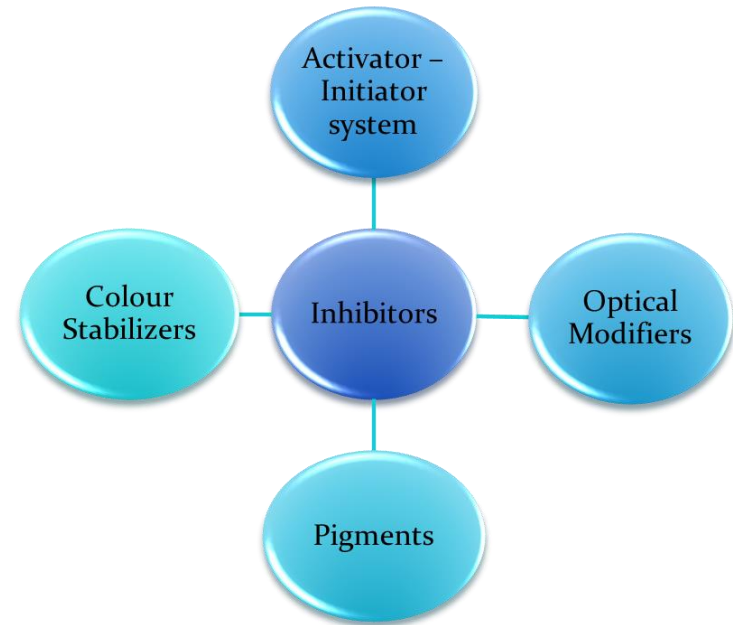
Reviewing last 60 years



COMPOSITION

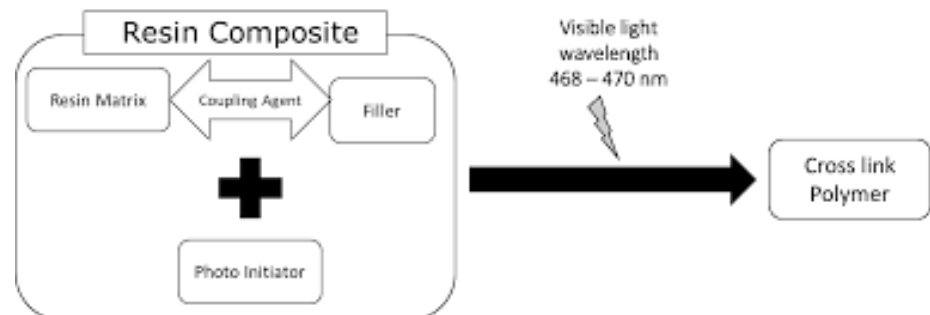


Minor Components

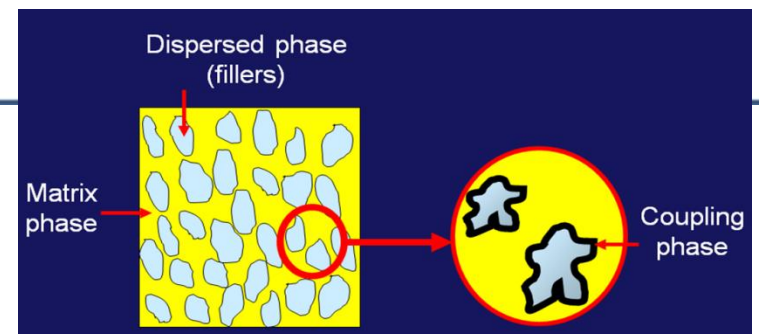


3 Major components

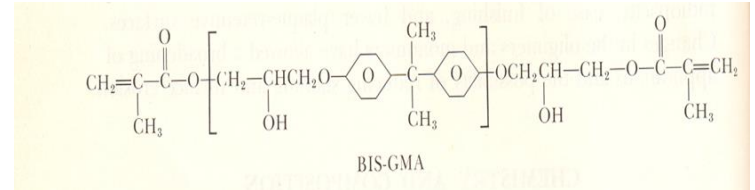
1. Resin matrix
2. Inorganic fillers
3. Coupling agents



- Basically, composite restorative materials consist of a **continuous polymeric or resin matrix** in which an **inorganic filler** is dispersed.



Resin matrix



*Principal monomers:

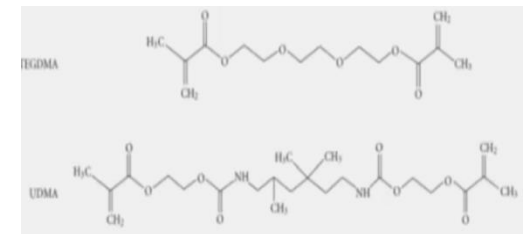
- **BIS-GMA**(Bis phenol A glycidyl methacrylate)"*Bowen's RESIN*"
- **UDMA** (Urethane di methacrylate)

Due To **High Density** of these Monomers Are **Viscous** ,so **diluent monomer** is added to **Decreases Viscosity** → **paste** like consistency

*Diluent monomers:

- **TEGDMA**(Triethylene glycol dimethacrylate)

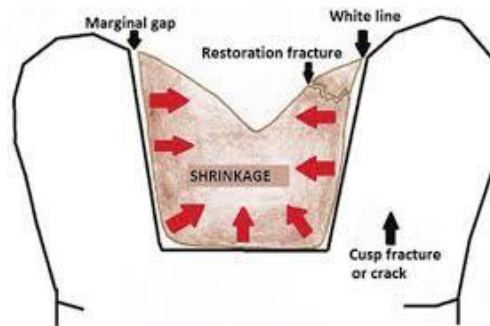
Forms highly cross linked polymer structures.



Inorganic Fillers

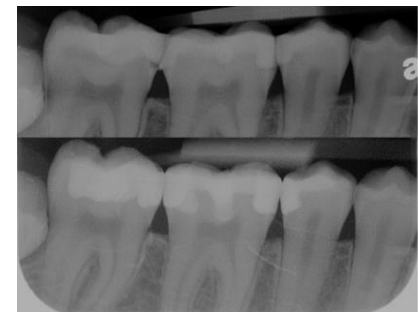
Commonly used fillers are

- Quartz (Used in early generation of composites, difficult to get good polish.)
- Silica particles
- Glasses
- Zirconium dioxide



Functions of fillers:

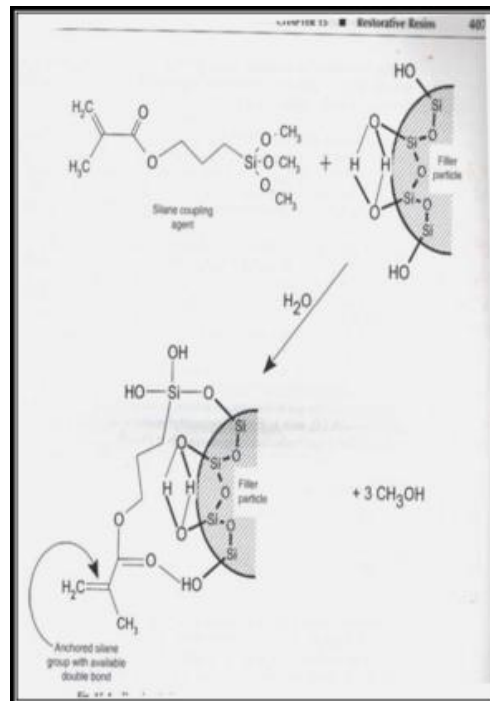
1. Decreases wear of composites
2. Increases hardness
3. Reduces polymerization shrinkage
4. Increases radiopacity
5. Contribute to esthetics



Coupling agents

It promotes adhesion between inorganic filler and organic resin matrix

- Organosilanes are most commonly used coupling agents

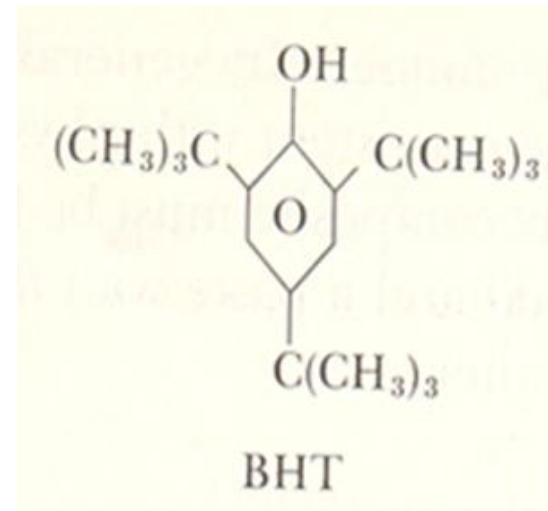


Inhibitors

Butylated Hydroxytoluene

FUNCTIONS:

1. Extend storage life
2. Increase working time



Optical Modifiers

- Titanium Dioxide
 - Aluminium Oxide (0.0001 TO 0.007 WT%)
- To achieve natural tooth like appearance



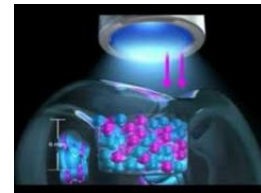
Activator initiator system



(Light activated composite)(Single paste system)

Photosensitizer - Camphorquinone

- Activator-visible blue light (Wavelength 400 - 500 nm)



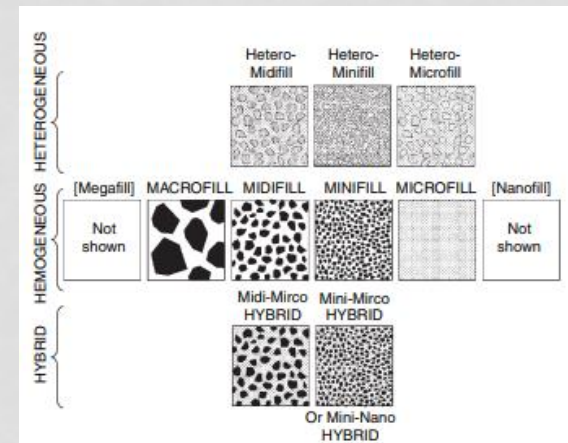
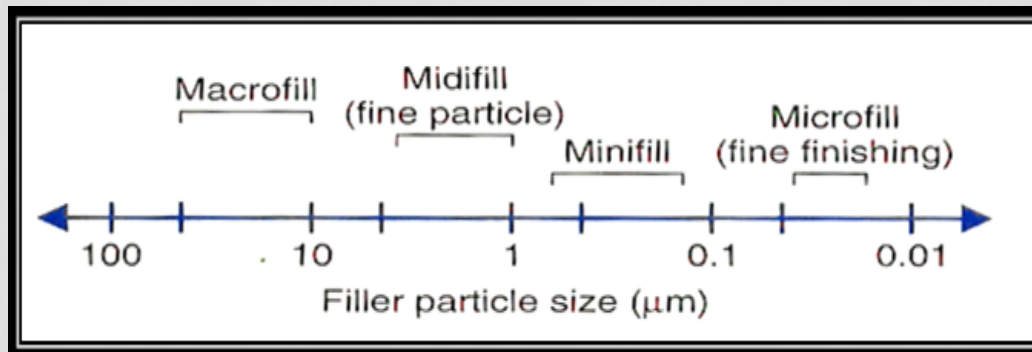
(Chemical activated composite)

- Initiator- Benzoyl peroxide
- Activator- Aromatic tertiary amine

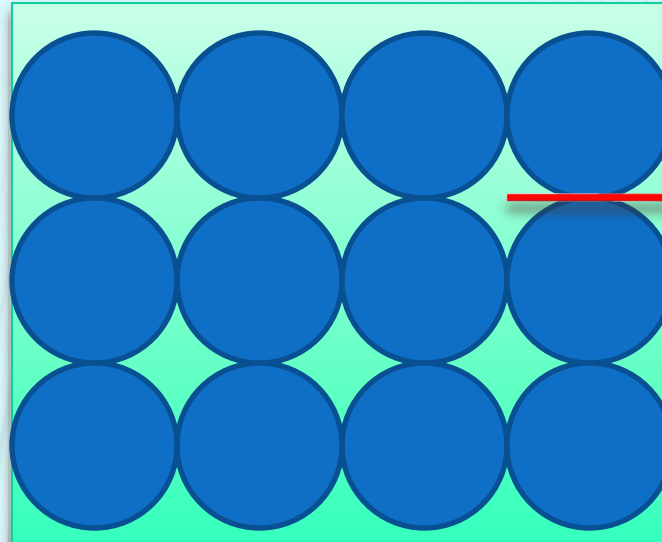


CLASSIFICATION OF COMPOSITE

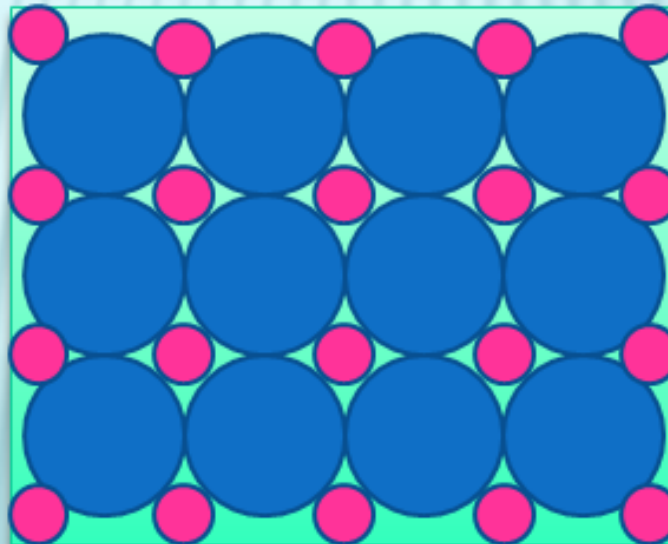
- According to Sturdevant, based on the **filler particle size**
 - ❖ **Megafill**- in this one or two large glass inserts 0.5 to 2 mm
 - ❖ **Macrofill**- Filler particle size range between 10 to 100 μm
 - ❖ **Midifill** - particle size range between 1 to 10 μm in diameter
 - ❖ **Minifill** - particle size range between 0.1 to 1 μm in diameter
 - ❖ **Microfill** - particle size range between 0.01 to 0.1 μm
 - ❖ **Nanofill** - particle size range between 0.005 to 0.01 μm



PARTICLES SIZE DISTRIBUTION



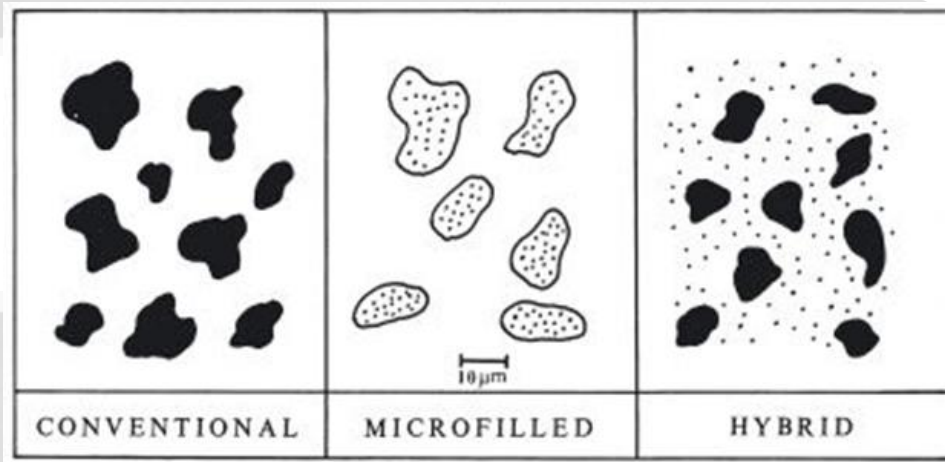
VOIDS/GAPS



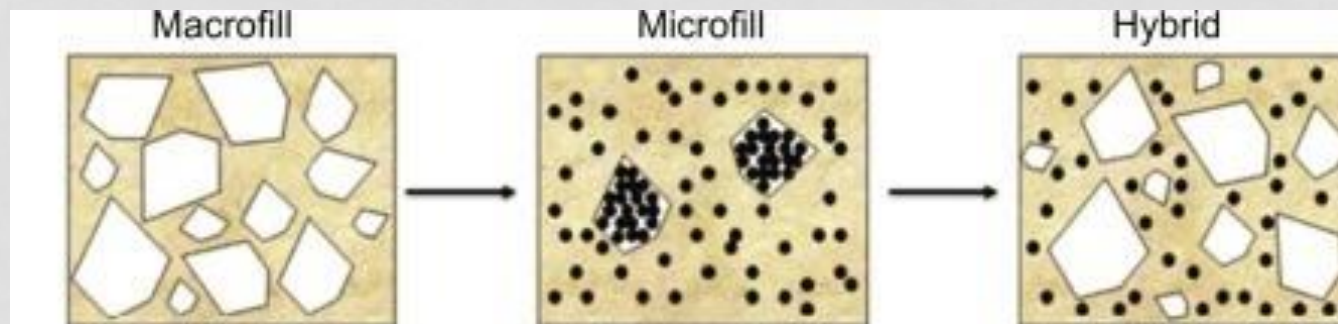
Effect of filler particle size



According to SKINNER'S



- Conventional composites (Macrofilled) 8-12μm
- Small particle filled composite – 1-5 μm
- Microfilled composite – 0.04 – 0.4 μm
- Hybrid composite – 0.6 – 1 μm



CONVENTIONAL / MACROFILLED COMPOSITES.

- **Filler size – 8 – 12 um**
 - **Indication: Can be Used in Class I and II restorations**

DISADVANTAGES :

1. **Difficult to polish**
2. **Surface roughness**
3. **Staining, plaque**
4. **Poor esthetics – *discolouration***



MICROFILLED COMPOSITES.

- Filler size – 0.04 to 0.4 μm
- Microfilled composites give restoration smooth surface and can be highly polished with the best esthetic results. Designed to give a smooth, lustrous surface similar to tooth enamel.

Indication :

- 1. Aesthetic restoration for anterior teeth(Class III)**
- 2. Restoring sub gingival areas**
- 3. Carious lesion on smooth surface (Class V)**

Advantage:

- 1. Best surface finish**

Disadvantage:

- 1. Less Tensile stress**
- 2. Less wear resistance**

HYBRID COMPOSITES

- Filler size – 0.4 to 1 μm

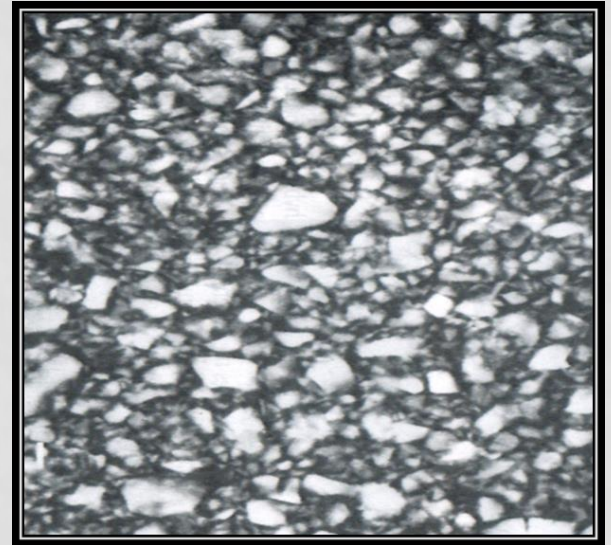
It is the **combination** of both Conventional and microfilled composite.(Combining the **favorable physical and mechanical properties of conventional** with the **smooth surface of microfill**).

Indication:

1. Anterior restoration (Class IV)
2. Non stress bearing area in posterior teeth

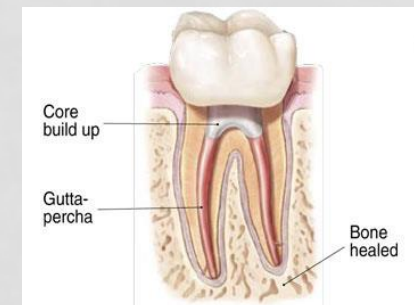
Advantage:

1. Good physical properties
2. Improve wear resistance
3. Good esthetics



OTHER VARIATIONS IN COMPOSITE

- Flowable composite
- Packable composite
- Laboratory composites
- Core composites





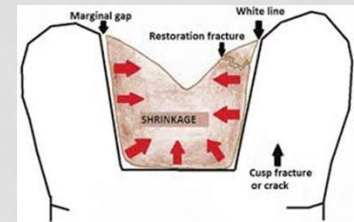
FLOWABLE COMPOSITES



- As the name indicates, it has low viscosity (more flow) their consistency is such that they can be flowed into the deep pit and fissure areas.
- Have **lower filler content**. Hence inferior physical properties (**Lower wear resistance and strength**) and exhibit high polymerization shrinkage.

Commonly used

- **Cervical lesions(Class V abfraction lesions)**
- **Preventive resin restorations**



PACKABLE COMPOSITES.

- Are **more viscous**(less flow)
- Feel upon insertion similar like amalgam.
- Have less polymerization shrinkage and low wear .
- Developed to accomplish **2 goals**.



1. Easier restoration of proximal contact.
2. Similarity to the handling properties of amalgam.

Recommended to use as class I,II restorations.

METHODS OF POLYMERIZATION



SELF
CURING

VISIBLE LIGHT
CURE

DUAL CURE
REQUIRES
BOTH SELF
CURE AND
LIGHT
CURE

SETTING REACTION

- The setting reaction for composites is called as Polymerization reaction
- The polymerized resin is highly cross-linked structure, because of presence of the difunctional carbon double bonds.

REFERENCES

- Craigs Restorative dental materials 11 th edition
Chapter 9 page 232- 237
- Art and science sturdevant operative dentistry 5 th
edition Page no.501